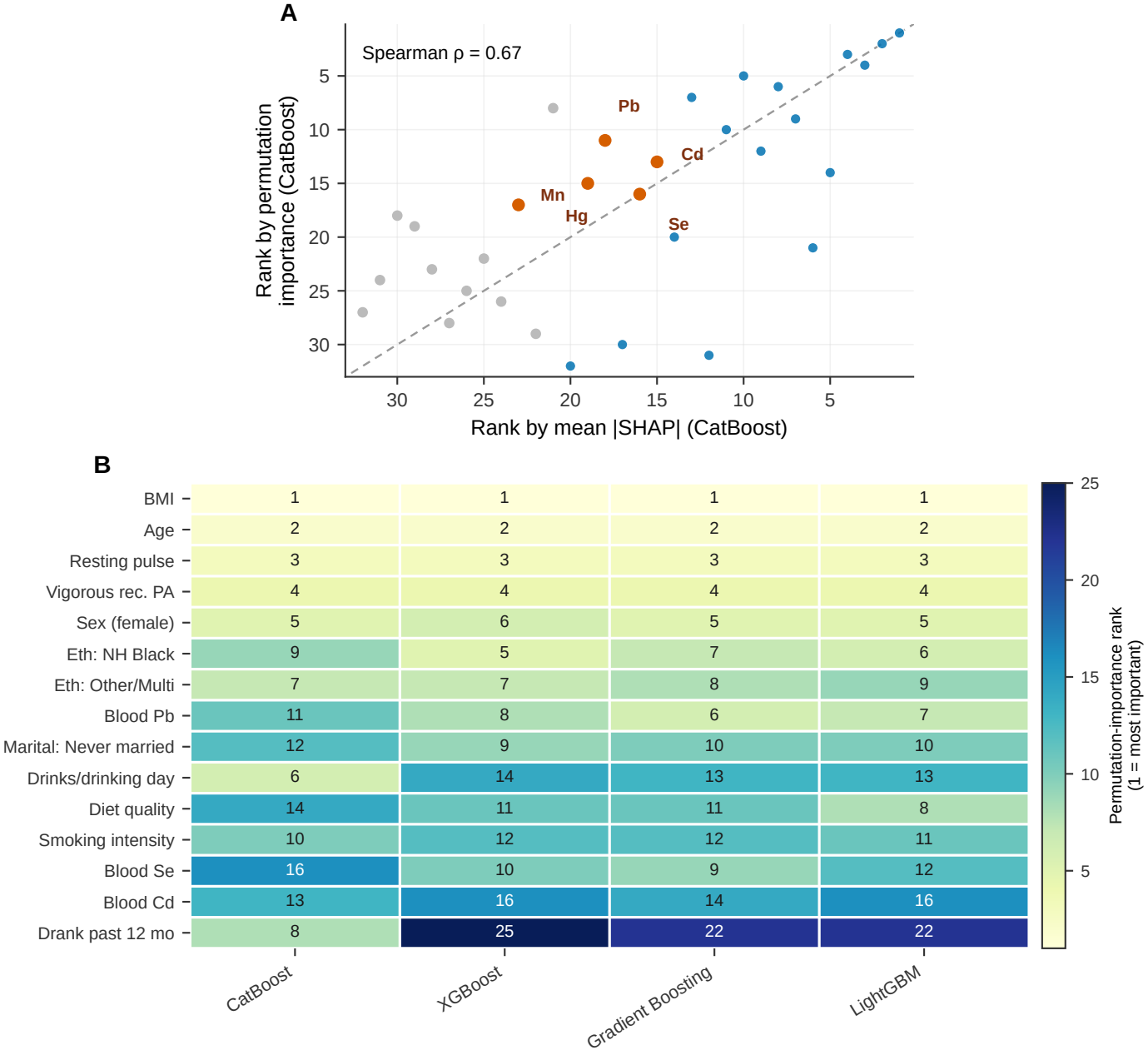
A
Spearman ρ = 0.67
Rank by permutation importance (CatBoost)
Rank by mean |SHAP| (CatBoost)
Pb
Cd
Mn
Hg
Se
5
10
15
20
25
30
B
BMI 1 1 1 1
Age 2 2 2 2
Resting pulse 3 3 3 3
Vigorous rec. PA 4 4 4 4
Sex (female) 5 6 5 5
Eth: NH Black 9 5 7 6
Eth: Other/Multi 7 7 8 9
Blood Pb 11 8 6 7
Marital: Never married 12 9 10 10
Drinks/drinking day 6 14 13 13
Diet quality 14 11 11 8
Smoking intensity 10 12 12 11
Blood Se 16 10 9 12
Blood Cd 13 16 14 16
Drank past 12 mo 8 25 22 22
CatBoost
XGBoost
Gradient Boosting
LightGBM
Permutation-importance rank
(1 = most important)
25
20
15
10
5